Positional Encoding Comparison

Reference: [Su et al., 2024] [Press et al., 2022] [Chen et al., 2023]

Absolute Positional Encoding

h = x + PE(pos)

Sinusoidal / Learnable vectors
Added directly to embeddings

□ Simple implementation

□ Cannot extrapolate to longer sequences

□ Performance drops beyond training length

Used by: Original Transformer

BERT, T5

Extrapolation: Poor

Current Status: Deprecated

RoPE (Rotary Positional Encoding)

Rotate Query and Key vectors

□□□ = f(|pos_i - pos_j|)
Depends only on relative position difference

□ Naturally supports relative position

□ Good extrapolation ability

□ Linear attention compatible

Used by: LLaMA 2/3

Qwen, Mistral

Extrapolation: Good

Current Status: Mainstream

ALiBi (Linear Bias)

□□□□□ += m × (j - i)

Does not modify embeddings
Added directly to attention matrix

□ Train 2K, infer 64K

□ Best extrapolation ability

□ Requires tuning slope m

Used by: MPT

BLOOM

Extrapolation: Best

Current Status: Niche